# aemon: Teaching a Small Model to Investigate Strategy Questions

Agentic distillation of theory-driven decision making into a 3B open model, with an executable environment and a frozen, blind evaluation

Hatem Saadallah Nour Jennane

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**Abstract**

Asked whether a manufacturer should take its technology to a new kind of customer, language models tend to produce consulting prose rather than an investigation. We teach a small open model, Tiny Aya Global (3.35B parameters), to carry out a theory-driven investigation (we call the resulting agent aemon; *weigh what is known, test what is not*): frame the decision, state rival causal theories, weigh quoted evidence for and against each, challenge its favourite, design a test with a decision rule or decline one, and revise on new evidence. The procedure follows the scientific approach to strategic decisions, which treats a theory as a restriction on beliefs and a decision as a choice between committing and experimenting. We build an executable environment with twelve guarded tools over 1,334 verified public records, a stage controller and executable checks; distil 3,891 tool-call rows from 158 replay-exact investigations by a Command A+ teacher into a LoRA adapter with AgentTune, plus 51 correction rows; verify the deployed GGUF adapter against the trained one; and evaluate on four companies never used for training or selection with two qualified, blinded LLM judges under a plan frozen before the run. The fine-tuned model scores 1.48 against 0.65 for the base model under its best prompt on a nine-dimension 0–3 rubric, a paired difference of +0.83 (95% case-cluster bootstrap interval $[0.69, 0.97]$), higher for every company and every kind of new evidence, and finishes 28 of 48 investigations against 4, at about $0.03 per investigation against $0.25 for Claude Opus. A final system, built after the frozen tests and measured on the same cases, adds a second adapter and decoding-time guards that remove failing actions: it finishes 46 of 48 investigations with 39 tool calls each (the base model needs 242) and scores +1.02 $[0.92, 1.10]$ over the base model. Claude Haiku 5.5, a closed model run through the same environment, scores higher still (2.28 against 1.62) for about $0.008 per investigation. A diagnosis traces the remaining failures (no declined tests, little directional new evidence, decision rules cut by a length cap) to gaps in the training data, and we report what environment changes and targeted examples did and did not fix.

## Contents

**1 Introduction** **4**

**2 Related work** **5**

**3 Theory: decisions as tests of theories** **5**
3.1 The scientific approach to strategy . . . . . 5
3.2 A theory as a restriction on beliefs . . . . . 5
3.3 From the framework to an agent . . . . . 6

**4 The investigation environment** **7**
4.1 An investigation as a sequential decision problem . . . . . 7

4.2 Tools . . . 7
4.3 Guards . . . 8
4.4 The stage controller . . . 9
4.5 Search . . . 9
4.6 Executable checks . . . 9
4.7 Cases and the dossier . . . 9

**5 The agent** **10**
5.1 Policy and observation . . . 10
5.2 Constrained decoding . . . 10
5.3 Sampling and stopping . . . 11

**6 Training data from a teacher** **11**
6.1 The teacher . . . 11
6.2 The replay filter . . . 11
6.3 Counts . . . 12
6.4 Correction rows . . . 12

**7 Training** **12**
7.1 Objective . . . 12
7.2 Parameter-efficient adaptation . . . 12
7.3 Procedure and settings . . . 13
7.4 The conversion gate . . . 13

**8 Evaluation design** **13**
8.1 What is measured . . . 13
8.2 Judges and their qualification . . . 14
8.3 Blinding . . . 14
8.4 The statistic . . . 14
8.5 Arms, selection and the freeze . . . 15
8.6 Mechanical audit . . . 15

**9 Results** **15**
9.1 Development . . . 15
9.2 The final test . . . 15
9.3 Cost . . . 17

**10 Diagnosis: what still fails, and why** **17**

**11 Iteration after the final test** **18**
11.1 Environment changes . . . 18
11.2 Targeted teacher examples . . . 19
11.3 A second test set . . . 19
11.4 The final system: a second adapter and adapter switching . . . 19

**12 What did not fix the remaining failures** **21**
12.1 Runtime interventions: the trained model does not use what it is told . . . 21
12.2 Training data: the teacher's errors are in the student . . . 22

**13 aemon: the agent as a product** **22**

**14 Limitations** **23**

**15 Conclusion** **24**

**A Reproducibility** **26**

# 1 Introduction

Ask a language model whether a manufacturer should take its technology to a new kind of customer, say whether a maker of professional laundry machines should build something for households, and it will usually answer with consulting prose: a list of factors, a confident market size with no source, and evidence collected only for the answer it started with. Good strategy work looks different. It frames the decision and its real alternatives, including doing nothing; it states rival theories of *who* would change their behaviour and *why*; it derives predictions that tell those theories apart; it weighs quoted evidence for and against each and looks hardest for evidence against its favourite; when information is worth its cost it proposes a test with a decision rule fixed in advance; and it revises when new evidence arrives [1, 2].

This paper asks whether that procedure can be *taught* to a small open model that runs on a single GPU, by distilling the tool-using investigations of a much larger model inside an environment that enforces the procedure's form. Our answer is a qualified yes. On a frozen test of four companies never used for training or selection, the fine-tuned 3.35-billion-parameter student (Tiny Aya Global with a LoRA adapter) scores 1.48 on a blind nine-dimension rubric against 0.65 for the same base model under its best prompt, a paired difference of +0.83 with a 95% case-cluster bootstrap interval of $[0.69, 0.97]$, and it finishes 28 of 48 investigations against 4. An investigation costs about $0.03 of rented GPU time, against $0.25 measured for Claude Opus on an earlier control run of the task. Claude Haiku 5.5, run on both test sets, does better than our final system for less (Section 9.3).

**Contributions.**

1. **An executable investigation environment** (Section 4): twelve guarded tools over a dossier of 1,334 verified public records when version 1 was trained and first tested (1,958 now), a stage controller that offers only the tools of the current step, and executable checks. Quotes must match a source word for word, theories must give distinct causal explanations, and a simulated what-if result may change a belief but can never support a recommendation about real demand.

2. **A distillation pipeline with provenance** (Sections 6 and 7): a Command A+ teacher runs the same environment; only investigations that replay exactly and reach an accepted finish become training rows, each bound by hash to its teacher episode and weights; training uses AgentTune's distillation trainer with loss on the target tool call only, followed by a conversion gate that checks the deployed GGUF adapter against the trained one.

3. **A pre-registered evaluation** (Sections 8 and 9): a held-out company split, two LLM judges each qualified on matched sound and flawed investigations, blinded packets with the key kept apart, a statistic and rule fixed before the run, and a final test run exactly once.

4. **An honest diagnosis and iteration** (Sections 10 and 11): where the agent still fails, why the training data explains it, and what environment changes and targeted teacher examples did and did not fix.

5. **A usable product** (Section 13): aemon, which runs the agent locally for people who are not analysts, with answer-first results, charts counted from the record, follow-up questions grounded in the investigation's sources, memory, attachments and PDF reports.

All code, the model and the data are public.[1]

[1] Model: https://huggingface.co/HatemSaadallah/aemon; data: https://huggingface.co/datasets/HatemSaadallah/aemon.

## 2 Related work

**Tool-using language agents.** Interleaving reasoning with actions [17] and teaching models to call tools [14] made language agents practical. Fine-tuning small models on agent trajectories from stronger models improves their tool use [3, 18]. Our setting differs in that the task has no single correct answer: quality lies in the *process* (framing, rival theories, evidence handling, experiment design, revision), so the environment must enforce form and the evaluation must judge substance.

**Distillation.** Knowledge distillation trains a small student on a larger teacher's outputs [8]. We distil at the level of tool calls: each training row is the context the teacher saw and the one call it made, with the loss on that call only, and with parameter-efficient adapters [9, 12] trained through AgentTune [10] on TRL [15].

**Structured generation.** Constraining decoding to a schema guarantees well-formed calls [16]. We constrain each turn to the tools open at the current step and to identifiers that exist, which removes a class of failures; we also found a cost, a length cap that cut fields mid-sentence (Section 10).

**LLM judges.** Strong models are useful but imperfect judges [19]. We qualify each judge on matched sound and flawed investigations before using it, blind every packet, and report both judges and their agreement.

**Strategy as science.** Camuffo et al. [1] found in a randomised trial with start-ups that founders taught to treat decisions as theory-driven experiments made better decisions; Camuffo et al. [2] formalise a theory as a restriction on beliefs. We take their procedure as the target behaviour (Section 3).

## 3 Theory: decisions as tests of theories

### 3.1 The scientific approach to strategy

Our target behaviour comes from the scientific approach to entrepreneurial and strategic decisions [1, 2]. A decision maker facing an uncertain opportunity should (i) frame the decision and its real alternatives, including doing nothing; (ii) state *theories*: causal accounts of who would change their behaviour and why; (iii) derive predictions that tell the theories apart; (iv) gather evidence for and against each, looking hardest for evidence against the favoured one; (v) when information is worth its cost, run a test with a decision rule fixed in advance; and (vi) revise when new evidence arrives. Field experiments with start-ups found that founders trained this way made more accurate decisions, pivoting or terminating sooner when evidence contradicted them [1].

### 3.2 A theory as a restriction on beliefs

Camuffo et al. [2] formalise a theory as a Bayesian network with a restriction. We follow their running example, Luxottica deciding whether eyeglasses could become a fashion product.

**Attributes and states.** An *attribute* is a question with an uncertain answer. Two attributes define the state space: can the firm design fashion eyewear ($Y$ or $N$), and will demand be high or low ($H$ or $L$). The four states are $\{YH, YL, NH, NL\}$, and the decision maker cares about the probability of high demand,

$$P(H) = \theta_Y \, \theta_{H|Y} + (1 - \theta_Y) \, \theta_{H|N}, \tag{1}$$

where $\theta_Y = P(Y)$, $\theta_{H|Y} = P(H \mid Y)$ and $\theta_{H|N} = P(H \mid N)$.

**The theory.** The theory "fashion design causes demand" is one arrow $Y \to H$ in a Bayesian network together with an inequality on its conditional probabilities,

$$\Theta = \{\theta : \theta_{H|Y} > \theta_{H|N}\}, \qquad \tilde{\Theta} = \{\theta : \theta_{H|Y} = \theta_{H|N}\}, \tag{2}$$

where $\tilde{\Theta}$ is the null hypothesis that design has no effect.

**Value of a theory.** The parameters are unknown, so the decision maker holds a distribution $\mu$ over them. Let $V_\Theta = \mathbb{E}_\mu[P(H) \mid \theta \in \Theta]$ and $V_{\tilde{\Theta}} = \mathbb{E}_\mu[P(H) \mid \theta \in \tilde{\Theta}]$. With prior $\omega$ that the theory is right, its value is

$$V = \omega\, V_\Theta + (1 - \omega)\, V_{\tilde{\Theta}}. \tag{3}$$

**A worked example.** Take $\theta_{H|N} = 0.5$ and $\theta_Y = 0.8$ with probability one, and $\theta_{H|Y}$ uniform. Under the theory $\theta_{H|Y}$ is uniform on $(0.5, 1)$, so $\mathbb{E}[\theta_{H|Y} \mid \Theta] = 0.75$ and

$$V_\Theta = 0.8 \cdot 0.75 + 0.2 \cdot 0.5 = 0.70. \tag{4}$$

Without the restriction $\theta_{H|Y}$ averages 0.5 and $P(H) = 0.5$. The theory raises belief in high demand by *ruling parameter values out*: it is informative precisely because it can be wrong.

**Experiments.** An experiment produces evidence that updates $\mu$, and with it $V_\Theta$ and $V_{\tilde{\Theta}}$. In their account the prior on the theory itself changes only when the evidence moves the beliefs about parameters far enough, $\|\mu' - \mu\| > \mu^*$: a small surprise adjusts parameters, a large one shakes belief in the theory. A rival theory has its own value $Q$, built the same way. With costly experiments the decision maker should commit once $V$ is far enough above or below $Q$, and experiment in between, the logic of optimal stopping for costly information [13]. Experiments are assumed Bayes plausible: before running one there is no reason to expect it to move the value in a particular direction.

**Remark 1 (Bayesian updating, illustrated)** *For intuition, suppose $n$ customers are shown the fashion design and $k$ buy. Under the theory, with $\theta_{H|Y} \sim U(\theta_{H|N}, 1)$, and under the null, with $\theta_{H|Y} = \theta_{H|N}$, the marginal likelihoods are*

$$m_\Theta(k) = \frac{1}{1 - \theta_{H|N}} \int_{\theta_{H|N}}^{1} \binom{n}{k} p^k (1-p)^{n-k}\, dp, \qquad m_{\tilde{\Theta}}(k) = \binom{n}{k} \theta_{H|N}^k (1 - \theta_{H|N})^{n-k}, \tag{5}$$

*and Bayes' rule gives the posterior on the theory*

$$\omega' = \frac{\omega\, m_\Theta(k)}{\omega\, m_\Theta(k) + (1 - \omega)\, m_{\tilde{\Theta}}(k)}. \tag{6}$$

*Full updating moves $\omega$ after every observation; the threshold rule of Camuffo et al. [2] moves it only after large surprises, so this calculation overstates how often a theory itself is revised.*

### 3.3 From the framework to an agent

The agent in this paper does *not* compute these quantities, and neither does its environment. A computed update would need an invented network and invented likelihoods, and assumed ratios could manufacture exactly the revision we want to measure. Instead the agent carries out the framework's steps in words, with tools and quoted evidence, and the environment enforces their form:

- *theories* are explicit causal claims (who benefits, why behaviour changes, how value is captured), and two of them must be genuinely distinct (Section 4);
- *evidence* is a quoted passage from a source, assessed as supporting, contradicting or having no bearing on a theory, with what it does not show;
- *tests* carry a prediction per theory and a decision rule, within a budget and a time limit;
- *beliefs* are stated on a five-step verbal scale (very unlikely, unlikely, uncertain, likely, very likely) when the agent takes a position and again after new evidence. The scale is a within-run signal, not a calibrated probability; a direction check asks only whether undermining evidence did not raise the affected theory and supporting evidence did not lower it.

The framework thus supplies the *grammar* of a good investigation; the environment makes that grammar executable; and fine-tuning teaches a small model to speak it.

# 4 The investigation environment

## 4.1 An investigation as a sequential decision problem

We model one investigation as a finite-horizon sequential decision problem. A *case* $c$ fixes a company, a question, a brief, budgets and a dossier $D_c$ of public records the case may see. The state $s_t$ holds everything on record at step $t$: the frame (objective, constraints, alternatives), the sources read, the theories, the assessed quotes, any test or declined test, the stated position, revisions and the final answer. An action is one tool call $a_t = (\mathit{name}, \mathit{arguments})$. A guarded transition $\mathcal{T}$ either accepts the call and updates the state, or refuses it with a reason and leaves the state unchanged:

$$(s_{t+1}, r_t) = \begin{cases} (\delta(s_t, a_t), \mathit{result}) & \text{if every guard on } a_t \text{ holds in } s_t, \\ (s_t, \mathit{refusal}(\mathit{category}, \mathit{reason})) & \text{otherwise.} \end{cases} \tag{7}$$

The agent never sees $s_t$ in full; it sees a compact observation $o_t = \phi(s_t)$ (Section 5). An episode ends when the agent finishes, after $E$ refusals in a row ($E = 4$ as evaluated), at a turn limit ($T = 45$), or when the context is full.

## 4.2 Tools

Staged runs expose twelve tools (Table 1). Each argument has an exact key set and length limits; a call with missing or extra keys is refused.

Table 1: The twelve tools, their arguments and what makes them refuse a call.

| Tool | Arguments | Refused when (category) |
|---|---|---|
| `read_brief` | none | arguments given |
| `frame_problem` | objective; 1–6 constraints; 2–6 alternatives | already framed (repeat); brief not read (order) |
| `search_evidence` | query | no frame yet (order); no searches left, or the last one is reserved for the challenge (budget); same search again (repeat) |
| `read_source` | id | the id never came from a search (fabricated) |
| `add_theory` | claim, who benefits, why behaviour changes, how value is captured; assumptions; predictions | more than 5 theories (budget); identical claim (repeat) |
| `assess_evidence` | theory, source, direction, quote, evidence kind, what it does not show | source not read (unread_evidence); quote not in the source (misquote); a simulated source used as support (synthetic_as_evidence); same theory–source pair (repeat) |
| `challenge_theory` | theory, query | unknown theory; budget; repeat |
| `propose_test` | name, $\geq$ 2 theories, method, sample, measure, one prediction per theory, decision rule, cost, weeks | over the test budget or time limit (budget); arguments |
| `decline_test` | reason | arguments |
| `state_position` | favoured theory, recommendation, one belief per theory | favoured theory never challenged (order) |
| `revise_theory` | theory, status (kept, revised, rejected), reason, new claim, evidence | cites unread evidence; a revised claim that does not change |
| `finish` | recommendation, target customer, value proposition, favoured theory, next action, uncertainty, what would change it, evidence; beliefs | cites unread evidence; cites a simulated record (synthetic_as_evidence) |

### 4.3 Guards

**Exact quotation.** An assessment's quote $q$ must appear in the source's excerpt $x$ after normalisation:

$$\mathit{valid}(q, x) \iff |q|_{\text{words}} \geq 3 \ \wedge \ |\nu(q)| \geq 15 \ \wedge \ \nu(q) \sqsubseteq \nu(x), \tag{8}$$

where $\nu$ lowercases, collapses whitespace and maps curly quotes, dashes and non-breaking spaces to plain forms, and $\sqsubseteq$ is substring containment. Surrounding quotation marks are stripped first: half of the teacher's early refused quotes were exact words inside quotation marks.

**Distinct theories.** Two theories are near-duplicates when the Jaccard similarity of their processed word sets (claim plus why behaviour changes) reaches $\tau = 0.6$:

$$J(A, B) = \frac{|A \cap B|}{|A \cup B|} \geq 0.6. \tag{9}$$

The tool refuses an identical claim; the executable check fails a run with any pair above the threshold.

**Simulated evidence.** A what-if record is written by the case authors and starts with "SIMULATED". It may be read and assessed as contradicting or neutral, and it may move a belief, but it may never support a theory or be cited in the recommendation.

### 4.4 The stage controller

A controller $\mathcal{C}$ maps the state to the current step and the tools open in it, $A(s_t)$, and adds a one-line instruction to every observation. Table 2 lists the steps and what moves each on. Decoding is constrained to $A(s_t)$, so a call to a closed tool cannot be produced at all.

Table 2: Steps of the stage controller.

| Step | Open tools | Moves on when |
|---|---|---|
| brief | read_brief | the brief is read |
| frame | frame_problem | a frame exists |
| evidence | search, read | 3 sources are read |
| theories | add_theory, search, read | at least 2 theories exist |
| assessment | assess_evidence, read, search, add_theory | every theory has a weighed source |
| challenge | challenge_theory | one search against a theory |
| challenge reading | read, challenge_theory | what the challenge found has been read |
| test | propose_test, decline_test, and the look-further tools | a test is proposed or declined |
| position | state_position, look-further tools | a position is stated (its favourite challenged) |
| new evidence | read | the what-if is read (cases with a what-if) |
| update | revise_theory | a revision cites the what-if |
| conclude | finish, revise_theory, look-further tools | the investigation finishes |

### 4.5 Search

Search is deliberately plain. Terms are lower-case alphanumeric tokens with stop words removed and a crude plural stripped. A record's score for a query $Q$ is the summed inverse document frequency of the shared terms,

$$\mathit{score}(r, Q) = \sum_{w \in Q \cap r} \log\Big(1 + \frac{N}{n_w}\Big), \tag{10}$$

with $N$ records and $n_w$ containing $w$; the top three are returned. Each case allows 8 searches, the last reserved for a challenge. Search knows nothing about theories, so finding evidence against a favourite is the agent's job.

### 4.6 Executable checks

Every finished or stopped run is checked mechanically. The checks include framing (at least three alternatives, including no expansion or a different route), distinct theories, causal form (no "should" in a claim), testable predictions, evidence use, evidence classes (no theory supported only by company claims or simulated results), evidence limits, disconfirmation (the favourite was challenged and what was found was read), experiment design (predictions that differ and a decision rule with a measurable threshold, or a decline where no test is warranted), revision on new evidence, the existing offering (premise records read), no invented numbers, no simulated evidence in the recommendation, a recommendation with a distinct next action and at least two sources, and at least 80% of calls accepted. The checks are guards against mechanical mistakes, not a measure of strategic quality; that is what the judges are for (Section 8).

### 4.7 Cases and the dossier

When version 1 was trained and first tested the dossier held 1,334 public records (935 independent observations, 399 company claims; 1,958 now, with the second test's companies and later training

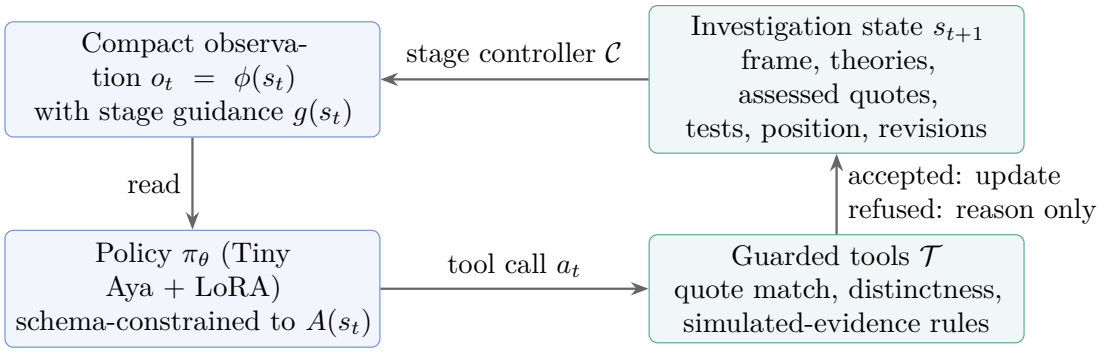



Figure 1: One step of an investigation. The policy reads a compact observation of the state, emits one tool call that decoding restricts to the open tools, and the guarded tools either update the state or refuse with a reason.

cases), each with source, URL, date and a short excerpt, every excerpt checked against its live source on the access date. Each case sees only its own company's records (and, for three original cases, a shared market set). There were then 137 development cases on 64 companies (207 on 99 now): for every company a base case and a variant in which an irrelevant simulated result arrives mid-investigation, and for three companies also supporting and undermining what-ifs. Every case allows 8 searches, a test budget of 20,000 euros and 12 weeks, which the brief calls analyst assumptions. Eighteen cases (nine companies) are marked as not warranting a test, so declining is the right call there.

The split is by company, never by row: 58 training companies (122 cases), 6 validation companies (15 cases, used for selection) and 4 test companies (16 cases: base, irrelevant, supporting and undermining for each of Aquaporin, Electrolux Professional, RATIONAL and Simonelli), held in a separate file that development runs never load.

# 5 The agent

## 5.1 Policy and observation

The policy $\pi_\theta$ is Tiny Aya Global [5] (3.349 billion parameters, 36 layers, hidden size 2,048) with a LoRA adapter. At each step it receives a fixed system prompt and one user message, the compact observation

$$o_t = \phi(s_t) \oplus g(s_t) \oplus \mathit{result}_{t-1}, \tag{11}$$

where $\phi$ serialises the state as it stands (the question, budgets, frame, unread search results, every source read with its excerpt, the theories, every assessment, the challenge, tests, position, revisions and the last twelve tried calls), $g$ is the controller's guidance (the step's instruction, its open tools and the searches left), and the last tool result is appended. Guidance is added to what the model is shown, never to the record, so every recorded episode replays exactly. The observation replaces a transcript: the model sees the state, not a history it could copy.

## 5.2 Constrained decoding

Each turn is decoded under a JSON schema $\Sigma(s_t)$ that admits exactly one call to a tool in $A(s_t)$, with identifiers drawn from those that exist (theory ids, sources read) and length limits per field. The schema guarantees a well-formed call; whether it is a *good* call is the model's. As evaluated, short text fields were limited to 240 characters, which we later found cut test designs mid-sentence (Section 10); the limits were raised to 900 for methods and decision rules and 500 for samples and measures afterwards.

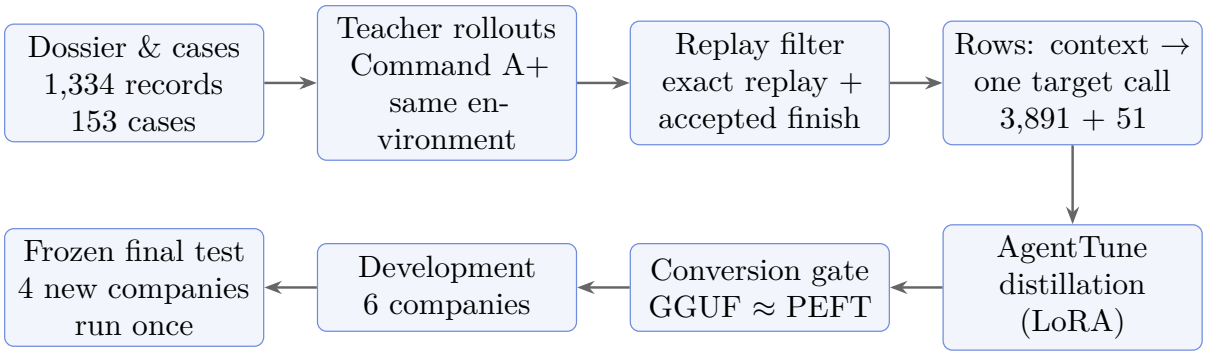



Figure 2: From dossier to evaluation. Only exact, accepted teacher investigations become rows; the trained adapter must pass the conversion gate before any evaluation; the final test runs once.

### 5.3 Sampling and stopping

Decoding is greedy (temperature 0). Straight after a refused call the next attempt samples at temperature 0.4 with a seed that depends on the case, the attempt and the number of refusals in a row,

$$seed = \sum_{ch \in case\ id} ord(ch) + 1000\, attempt + errors, \tag{12}$$

so a retry is not a repeat of the refused call. A run stops after 4 refusals in a row or 45 turns, with a context of 12,288 tokens and 520 tokens per turn.

**Algorithm 1.** One investigation (the evaluated runner)

**input:** case $c$, policy $\pi_\theta$, prompt $P$; **limits:** $T = 45$, $E = 4$
$s \leftarrow s_0(c)$; $e \leftarrow 0$; $o \leftarrow$ "A new investigation is ready. Call read_brief."
**for** $t = 1, \ldots, T$ **do**
  $\tau \leftarrow 0$ **if** $e = 0$ **else** 0.4
  $a \leftarrow \textsc{Decode}(\pi_\theta, P, o;\ \Sigma(s), \tau, seed(c, e))$ (schema-constrained)
  $(s', r) \leftarrow \mathcal{T}(s, a)$
  **if** $r$ is a refusal **then** $e \leftarrow e + 1$ **else** $e \leftarrow 0$; $s \leftarrow s'$
  **if** $s$ is finished **then return** $s$, "finished"
  **if** $e \geq E$ **then return** $s$, "repeated errors"
  $o \leftarrow \phi(s) \oplus g(s) \oplus r$
**return** $s$, "turn limit"

## 6 Training data from a teacher

### 6.1 The teacher

The teacher is Command A+ [4] (Apache 2.0), quantised to Q4_K_M and served with llama.cpp [7] on a single A100, with the same tools, controller and checks as the student and a teacher prompt that adds rules for good practice. Claude was excluded as a teacher because its commercial terms bar using its outputs to train a competing model; Aya Expanse 32B was tried first and rejected for weak reasoning.

### 6.2 The replay filter

A teacher investigation becomes training data only if it is *exactly reproducible* and *accepted*: replaying its recorded turns through a fresh environment must reproduce every tool result and the recorded call sequence, and must reach an accepted finish. Its provenance must match the permitted teacher weights in an append-only manifest. Passing the executable checks is recorded

but not required, because a teacher investigation can be instructive while failing a check; the checks are reported, and the judges score quality.

**Algorithm 2.** Rows from teacher investigations

**for each** teacher episode $\epsilon = (c, \{(a_t, r_t)\}_{t \le T})$ **do**
  **require** $hash(weights(\epsilon)) \in manifest$ **and** $\epsilon$ not excluded on review
  $s \leftarrow s_0(c)$
  **for** $t = 1, \dots, T$ **do**
    $(s', r') \leftarrow \mathcal{T}(s, a_t)$; **require** $r' = r_t$ (exact replay)
    **if** $a_t$ was accepted **then** emit row $\big(x = P \oplus \phi(s) \oplus g(s) \oplus r_{t-1},\ y = a_t\big)$
    $s \leftarrow s'$
  **require** $s$ finished and the finish accepted; **else** discard every row of $\epsilon$

Refused calls never become targets. Each row is bound by hash to its teacher episode and weights. A row longer than 8,192 tokens is refused, never truncated; the longest kept row is 6,011 tokens.

### 6.3 Counts

174 teacher investigations passed the filter, giving 4,302 rows. Training used the 3,891 rows from 158 investigations on 57 training companies; the 411 rows on validation companies were kept for held-out loss only.

### 6.4 Correction rows

To teach recovery from loops, we harvested states from the student's own runs on training companies where a loop began (a refused repeat, a third read of one source, or two refusals in a row; at most three per run) and asked the teacher for the single best next call, preceded by what had been learned, what remained uncertain and why the call helps. A reply was kept only if it was valid JSON, did not repeat an earlier call, was accepted by the tools, named something from the state, and acted on what it named; in the test step it had to propose a test with distinct predictions and a numeric rule or decline with a reason. 80 replies were provisionally kept and 51 became rows (49 at the theories step, one assessment, one conclusion), 0.35% of supervised tokens.

## 7 Training

### 7.1 Objective

Each row is a context $x$ and a target call $y = (y_1, \dots, y_m)$. The student is trained by supervised distillation with the loss on the target tokens only:

$$\mathcal{L}(\theta) = -\frac{1}{\sum_i m_i} \sum_i \sum_{j=1}^{m_i} \log \pi_\theta(y_j^{(i)} \mid x^{(i)}, y_{<j}^{(i)}), \tag{13}$$

implemented by giving every context token the label $-100$; the context's tokens must be an exact prefix of the full row's tokens, or the build fails. This is sequence-level distillation from the teacher's chosen calls [8].

### 7.2 Parameter-efficient adaptation

Every linear layer $W_0 \in \mathbb{R}^{d \times k}$ in attention and MLP blocks is adapted with a low-rank update [9],

$$W = W_0 + \frac{\alpha}{r} BA, \qquad B \in \mathbb{R}^{d \times r},\ A \in \mathbb{R}^{r \times k}, \tag{14}$$

with rank $r = 16$, $\alpha = 32$ and dropout 0.05; $W_0$ is frozen. The adapter has 504 tensors (about 121 MB).

### 7.3 Procedure and settings

Training used AgentTune's distillation trainer [10] with a TRL supervised trainer [15] and PEFT [12]. Settings: 2 epochs; learning rate $1 \times 10^{-4}$ with 5% linear warmup and cosine decay; batch size 1 with gradient accumulation 4 (effective batch 4); maximum length 8,192 tokens; bf16; seed 7. On the 3,942 training rows (3,891 teacher rows and 51 corrections) this is $3942 \times 2/4 = 1{,}971$ optimiser steps. Validation loss is the unweighted mean over validation companies of up to 40 evenly spaced rows each, so no company dominates; the best checkpoint by validation loss is kept. An overfit gate runs first: 10 rows for 60 steps must be reproduced greedily in at least 8 of 10 cases, or training stops. Training ran on a 40 GB slice of an A100 on a university cluster. The final held-out loss was 0.7125, at the last checkpoint.

### 7.4 The conversion gate

The trained PEFT adapter is converted to GGUF for llama.cpp, and a conversion can silently change behaviour. Before any evaluation, the gate replays recorded points of real investigations and compares, over the full vocabulary of 262,144 tokens, the converted adapter's effect on the next-token distribution with the PEFT adapter's effect,

$$\Delta_{\mathrm{PEFT}} = \log p_{\mathrm{base+LoRA}} - \log p_{\mathrm{base}}, \qquad \Delta_{\mathrm{GGUF}} = \log q_{\mathrm{base+LoRA}} - \log q_{\mathrm{base}}, \tag{15}$$

passing when the cosine similarity of the two effects is at least 0.99 and the probability-weighted residual RMS is at most 0.15 at every point, and a tensor audit confirms the factor pairs were cast as expected. Comparing *effects* rather than absolute log-probabilities matters: the base model alone differs by up to 0.19 between the two backends, which made an earlier absolute-gap gate fail for reasons that had nothing to do with the adapter. The final adapter passed on 11 points with minimum cosine 0.9990 and maximum residual 0.080; the file evaluated is byte-identical to the one gated (sha256 41794b57…).

## 8 Evaluation design

### 8.1 What is measured

The primary outcome is *investigation quality* as judged blind against a frozen rubric of nine dimensions, each scored 0 to 3 (Table 3). A judge gives the highest rung whose conditions are all met; an invented number, customer or company intention caps that dimension at 1. Dimensions the run never reached score 0; dimensions that do not apply (revision when no new evidence arrived) are excluded. An episode's score is the mean over its scored dimensions.

Table 3: The rubric (abridged). Scores 0 to 3.

| Dimension | 2 | 3 |
|---|---|---|
| Problem framing | separates the objective from the suggested solution | also finds an option or constraint a good analyst would not want missed, or corrects a false premise |
| Causal mechanism | who changes behaviour and why | the full chain from behaviour change to value this manufacturer captures |
| Distinct theories | rival explanations of the same question | rivals implying different products, customers or channels |
| Assumptions and predictions | observable | observable, and the theory is in trouble if they fail |
| Evidence judgement | separates company assertions, observations, studies and analogies | says what each key source cannot establish |
| Seeking challenge | genuinely searches against its favourite | reports fairly what it found, including nothing |
| Experiment design | separates at least two theories, or soundly declines | also feasible and decision-relevant |
| Revision | a proportionate, evidence-linked response to the affected theory | also traces the implications without overreacting |
| Recommendation | follows from the evidence, with stated uncertainty | commits to a next action and says what would reverse it |

### 8.2 Judges and their qualification

Two LLM judges score every packet: Claude (Opus 5.5, 8,000 output tokens, medium effort) and Codex (GPT-6-astra through its command-line tool, with shell, web, memory and repository access disabled). Before use, each was *qualified* on 20 matched pairs of fictional investigations, one sound and one deliberately flawed in each pair, under a rule registered before it was computed: the sound version must score strictly higher in at least 17 of 20 pairs, with ties and invalid outputs counted as failures, and at most 4 of 40 outputs invalid after one retry. Claude passed 20 of 20 at 8,000 tokens (19 of 20 at 3,000, with one truncated output, which led to the higher limit); Codex passed 20 of 20. A caveat: Codex wrote the qualification packets and their key.

### 8.3 Blinding

A packet shows the investigation as a reader would see it, with no arm, model, adapter, prompt or run name; refused calls are omitted and identical consecutive calls are shown once with a count. Packet ids are content hashes, the order is shuffled with a fixed seed (20261106), and the key linking packets to arms is kept outside the repository while judging.

### 8.4 The statistic

Let $q_{j,k}(c,a)$ be judge $j$'s score of arm $k$ on case $c$, attempt $a$. The paired difference for one case and attempt, averaged over the two judges, is

$$d(c,a) = \frac{1}{2}\sum_{j=1}^{2}\left[q_{j,\text{trained}}(c,a) - q_{j,\text{base}}(c,a)\right], \qquad \bar{d} = \frac{1}{|\mathcal{P}|}\sum_{(c,a)\in\mathcal{P}} d(c,a), \tag{16}$$

over the set $\mathcal{P}$ of pairs both arms produced. The 95% interval is a case-cluster percentile bootstrap [6]: in each of $B = 4000$ resamples, cases (not runs) are drawn with replacement with all their attempts, $\bar{d}^{*}_{b}$ is recomputed, and the interval runs from the 101st to the 3,900th of the sorted resamples (seed 20261106). Resampling whole cases respects the fact that three attempts on one case are not independent. Agreement between judges is the Spearman correlation of episode scores.

### 8.5 Arms, selection and the freeze

The comparison is the same base model with and without the adapter, on the same cases, controller, settings and runtime. The base runs under its best prompt (the staged prompt, chosen on validation companies: it beat the compact prompt by +0.30 there); the fine-tuned model runs under the compact prompt its rows were built with. Model selection used only the six validation companies: the higher mean blind-judge score wins, completion breaks a tie.

The final test was frozen before it ran: 16 held-out cases, 3 attempts each (attempts 0 to 2), both arms, 96 runs; the runner code, the job script, the llama.cpp build, context 12,288, 520 tokens per turn, 45 turns, 4 refusals in a row, temperature 0 with 0.4 only on a retry; the judges and their configuration; the statistic. Only infrastructure retries were allowed, never on model behaviour. It ran once, on 7 November 2026, and was judged on 8 November.

### 8.6 Mechanical audit

Every run is also replayed and scored with AuditKIT [11]: a recorded-mode evaluation of each episode, built from AgentTune event logs, with tool-call validity, redundant tool calls, loop detection and tool permission scorers, a full replay that must match the record, and a custom predicate for our guards. Every final-test run replayed with zero mismatches.

## 9 Results

### 9.1 Development

On the six validation companies (15 cases, one attempt each; 60 runs judged), the adapter chosen as final scored 1.29 (mean of judges) against 0.43 for the base under its best prompt, a paired difference of +0.86 $[0.61, 1.11]$, and finished 7 of 15 runs against none (Table 4). It beat the previous adapter by +0.20 $[-0.09, 0.53]$, an interval that includes zero; it was chosen under the registered rule (higher mean judge score on validation).

Table 4: Development results (6 validation companies, 15 cases, 1 attempt).

| Arm | Finished | Claude | Codex | Mean of judges |
|---|---|---|---|---|
| Base, compact prompt | 0/15 | 0.12 | 0.16 | 0.14 |
| Base, staged prompt (best) | 0/15 | 0.35 | 0.52 | 0.43 |
| Second adapter | 6/15 | 0.88 | 1.31 | 1.09 |
| Final adapter | 7/15 | 1.09 | 1.49 | 1.29 |

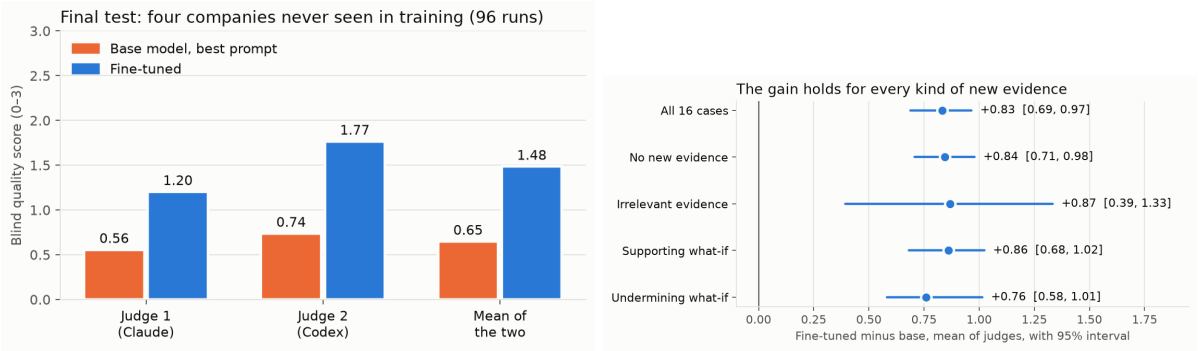
### 9.2 The final test

Table 5 gives the frozen final test: 16 held-out cases on four companies, three attempts, 96 runs.

Table 5: Final test (4 held-out companies, 16 cases, 3 attempts, run once). Differences are paired by case and attempt with 95% case-cluster bootstrap intervals.

| | Base (best prompt) | Fine-tuned | Difference [95% CI] |
|---|---|---|---|
| Judge 1 (Claude) | 0.56 | 1.20 | +0.64 [0.51, 0.77] |
| Judge 2 (Codex) | 0.74 | 1.77 | +1.03 [0.87, 1.18] |
| **Mean of the two judges** | **0.65** | **1.48** | **+0.83 [0.69, 0.97]** |
| Finished with a recommendation | 4 of 48 | 28 of 48 | |
| Stopped: repeated refused calls | 19 | 11 | |
| Stopped: context full | 25 | 0 | |
| Stopped: turn limit | 0 | 9 | |
| Refused calls (all runs) | 279 | 242 | |
| Mean calls per run | 20.2 | 35.4 | |
| Recovered after a refusal | 7 | 55 | |
| Reached the what-if (of 36) | 11 | 28 | |
| AuditKIT replay mismatches | 0 | 0 | |

The gain holds for every company (mean of judges: Aquaporin +1.10, Simonelli +0.98, RATIONAL +0.66, Electrolux Professional +0.60) and every kind of new evidence (none +0.84 [0.71, 0.98], irrelevant +0.87 [0.39, 1.33], supporting +0.86 [0.68, 1.02], undermining +0.76 [0.58, 1.01]). The judges rank the investigations alike (Spearman 0.94 over 95 episodes scored by both; one base output from Claude was invalid and is unpaired), although Codex scores both arms higher. The result survives two checks a reader may ask for (Section 14): resampling whole companies instead of cases, and comparing only the rubric dimensions both runs reached.




(a) Quality by judge.

(b) Gain by kind of new evidence.

Figure 3: Final test. Left: mean blind score per arm and judge. Right: fine-tuned minus base by kind of new evidence, with 95% intervals.

**By dimension.** Table 6 shows that the largest gains are in framing, causal mechanism and distinct theories, the parts of the procedure the environment shapes most directly. Evidence judgement, experiment design, revision and the recommendation improve but remain the weakest.

Table 6: Mean score per dimension, base → fine-tuned.

| Dimension | Claude | Codex |
|---|---|---|
| Problem framing | 1.00 → 1.63 | 1.00 → 2.90 |
| Causal mechanism | 0.92 → 1.44 | 1.15 → 2.63 |
| Distinct theories | 0.57 → 1.60 | 0.98 → 2.56 |
| Assumptions and predictions | 0.92 → 1.73 | 1.27 → 2.13 |
| Evidence judgement | 0.17 → 0.79 | 0.40 → 1.17 |
| Seeking challenge | 0.62 → 1.04 | 1.13 → 1.79 |
| Experiment design | 0.40 → 0.98 | 0.40 → 1.08 |
| Revision | 0.23 → 0.94 | 0.03 → 0.81 |
| Recommendation | 0.09 → 0.58 | 0.08 → 0.58 |

**Beliefs.** Of the fine-tuned runs that stated a belief about the affected theory both before and after a what-if, 6 of 6 moved in the allowed direction (3 undermining, 3 supporting) and none the wrong way; two runs lacked one of the two statements. The base model stated no such pair.

### 9.3 Cost

A fine-tuned investigation reads about 121,700 prompt tokens and writes about 2,900, in 66 seconds on one A100 slice (54 investigations per GPU hour). At the October 2026 median rental price of $1.76 per hour that is $0.032 per investigation ($0.016 at the lowest price, $0.068 at a hyperscaler's). A measured control of Claude Opus 5.5 on the same task, run earlier on the first harness and the original cases (15 investigations), cost $0.248 per investigation, and the same token counts at Opus list prices would cost $0.544. The comparison is of cost, not quality: Opus was not run on the final test.

**A closed small model: Claude Haiku 5.5.** Registered before it ran, Claude Haiku 5.5 (effort medium) was run on both test sets through the same environment, tools, stage controller and limits, with the base model's best prompt, three attempts per case, and judged blind with the base model, version 1 and the final system (Section 11.4) by the same two judges. Runs lost to API errors (29 of 120) were rerun; five Haiku runs lack a Claude Code judgment and are left out of the pairs.

| Mean of judges | Base | Version 1 | Final system | Haiku 5.5 | Final system − Haiku |
|---|---|---|---|---|---|
| Final test (48 runs) | 0.61 | 1.45 | 1.62 | 2.28 | −0.67 [−0.79, −0.52] |
| Test set 2 (72 runs) | 0.56 | 1.07 | 1.67 | 2.30 | −0.63 [−0.75, −0.52] |

Haiku scores higher at every company and on all nine dimensions, most on the recommendation and the causal mechanism; Codex alone puts the gap at −0.43 on both sets, the Claude Code judge at −0.90 and −0.86. It finishes 44 of 48 and 63 of 72 runs with 24 tool calls per finished investigation, at about $0.008 per investigation, a quarter of the self-hosted cost. Fine-tuning takes the 3B model about 60% of the way from its base to Haiku; what it offers over the closed model is not quality or price but local, offline use with open weights.

## 10 Diagnosis: what still fails, and why

**Unfinished runs.** 20 of the fine-tuned model's 48 final-test runs ended before a recommendation. Nine reached the turn limit, eight of them rereading the same few sources (19 to 30 reads of 4 to 10 sources). Eleven stopped after repeated refusals, seven of them finish calls that cited only the simulated what-if, which the rules forbid (an eighth such run reached the turn limit), and the rest repeated searches and assessments.

**Finished runs.** Of the 28 finished runs, the experiment check fails in 23. In 17 of them the case says no test is warranted, so the right call was to decline, but the model proposed a test; in 6 the check read the predictions as too similar, and in 5 of those 6 the predictions were in fact opposite comparisons that the check misreads (a known flaw). Evidence classes fail in 13 and evidence kinds in 11.

**The training data explains much of it.**

- No training row targets `decline_test`: in all 15 training cases with no test warranted that the teacher ran, the teacher proposed a test anyway. The student cannot imitate a decision it never saw.
- Directional what-if evidence is rare: 6 undermining and 2 supporting among 205 teacher investigations on training companies, against 103 irrelevant and 94 with none.
- Constrained generation capped short text fields at 240 characters and cut them mid-sentence: 132 of the 158 training test designs and 23 of the 37 accepted final-test proposals have a decision rule stopped at the cap. The model was trained on incomplete decision rules.
- On Aquaporin, no run in either arm read the company's insolvency records, and all eleven finished fine-tuned runs recommend investing: searching for a company's current position before theorising about it is a missing habit.

**Teacher and student fail differently.** The student fails the experiment check far more often than the teacher (82% against 39%), and evidence classes more (46% against 27%); it fails invented numbers (21% against 70% of the 223 finished teacher runs) and evidence kinds (39% against 83%) less. The case distributions differ, so this is not a matched comparison, but it suggests the student learned the teacher's form better than some of its judgement.

# 11 Iteration after the final test

Everything in this section is development work, reported separately from the final test, which is unchanged.

## 11.1 Environment changes

We tried changes to the environment alone, on the six validation companies with three attempts each (45 runs per configuration), under a rule written before the runs: a configuration replaces the current one only if it finishes more runs and does not lower the mean confidence score (Section 13).

Table 7: Environment configurations on development (45 runs each).

| Configuration | Finished | Refusal stops | Turn limit | Misquotes | Confidence |
|---|---|---|---|---|---|
| As evaluated | 24 | 17 | 4 | 35 | 2.21 |
| Reread limit, dates, help, 60 turns | 23 | 22 | 0 | 7 | 2.17 |

Refusing a third read of the same source removed the turn-limit loops but turned them into refused calls, which end a run after four in a row, and help after two refusals did not rescue enough of them, so the configuration was not adopted. Misquotes fell from 35 to 7; showing each source's date in the state is the likely cause, though it was not tested separately. Two more were registered. One requiring three weighed sources per theory finished 25 of 45 with confidence 3.12

and replaced the evaluated settings in the Strategy Lab, partly by construction, since one of the five confidence checks counts three weighed sources; one with source dates and stuck help but no reread limit was cancelled to free the GPUs for test set 2 and not rerun. The frozen tests keep the evaluated settings.

### 11.2 Targeted teacher examples

We then tried to fix the data gaps directly with targeted examples: states harvested from the student's own runs at the test step, at rereading loops and at refused finishes, answered once by the teacher under instructions for the missing skill, filtered by code gates and reviewed by a human before any could train. Three advisory reviews (by separate language-model agents, each with a different lens, and by Gemini) found the first rounds' answers to be templates: test arms that differed only in a message, rival theories the answer itself called unsupported, decision rules that jumped to developing a product line, and facts attributed to a case's year from later sources. The gates were tightened (complete fields, no message-only arms without a control, a bounded next step, the currency and sources of the case's time) and, after the human review, agree with all six of the reviewer's decisions on the second round; of those six answers, two were kept. The later rounds, as corrective distillation, are reported in Section 12.

### 11.3 A second test set

Because the four final-test companies have informed the diagnosis, a new claim needed new companies. A second held-out set of twelve companies and 24 present-day cases was built with the same pipeline, frozen before any run, and run once under the unchanged protocol. On the mean of the two judges the fine-tuned model scores 1.09 and the base model 0.57, a paired difference of +0.52 [0.33, 0.70] with a case-cluster bootstrap (Claude +0.41 [0.24, 0.57], Codex +0.63 [0.43, 0.84]; Spearman 0.95 between the judges over all 144 runs). The fine-tuned model is higher for 10 of 12 companies, and it finishes 32 of 72 investigations against 2 of 72. The gain is smaller than on the first test (+0.83 there).

### 11.4 The final system: a second adapter and adapter switching

The final system adds two parts to version 1. Both were built after the two frozen results were known, and the decoding-time guards below were designed from failures in the final-test runs, so none of these comparisons is held out: version 1's results above are the run-once result. Version 3 continues version 1 on rows from a second teacher (DeepSeek V4-Pro, 3,611 rows from 144 replay-exact investigations). *Dig then reason* uses one server holding both adapters: version 1 writes the brief, framing and evidence-gathering turns, version 3 every later turn. Both were judged blind together with the base model and version 1 by two judges qualified for this purpose (Claude Code and Codex on `gpt-6-sol`; mean of the two; case-cluster bootstrap), in the five-arm re-judging that also holds the combined system, which moves each arm by up to 0.02 against the three-arm re-judging:

| Arm | Test set 2 (72 runs) | Final test (48 runs) |
|---|---|---|
| Base, staged prompt | 0.56 | 0.61 |
| Version 1 | 1.07 | 1.45 |
| Version 3 | 1.48 | 1.35 |
| Dig then reason | 1.43 | 1.44 |

On test set 2 dig then reason is clearly above version 1 (+0.36 [0.22, 0.51]); on the final test it is level with it (−0.01 [−0.20, 0.18]), and version 3 alone is not distinguishable from version 1 either (−0.10 [−0.27, 0.05]). Version 1 remains the evaluated model; dig then reason is the most consistent later system and, with the loop guards below, the app's default.

**Loop guards.** Most refused calls in the final-test runs of dig then reason were repeats (233 of 274: the same search again, or a source already weighed for that theory), and four refusals in a row end a run. Three guards act only at decoding, with no retraining and no environment check changed: the assessment tool offers, per theory, only sources not yet weighed; a quote is chosen from the source's own sentences; and a call refused as an exact repeat is sampled again (at most three times, every refused draw kept in the record). With the step-budget aid of Section 12 added, judged blind by both judges against dig then reason as evaluated:

| | Development (45 runs) | Test set 2 (72 runs) |
|---|---|---|
| Finished runs, as evaluated → guarded | 28 → 45 | 43 → 63 |
| Refused calls per run | 5.2 → 3.1 | 6.1 → 3.5 |
| Quality, as evaluated → guarded | 1.42 → 1.62 | 1.43 → 1.59 |
| Difference [95% interval] | +0.19 [0.10, 0.30] | +0.16 [0.05, 0.27] |

The gain holds for each judge alone on both sets. On test set 2 the budget aid adds nothing over the three guards (+0.006 [−0.010, 0.022]), and one company (Testo) scores lower with the guards under both judges. These runs also exposed a bug in the quote guard: it offered a dateline that the environment's quote rule refuses, which stopped all six Belimo runs of each guarded arm. With the guard restricted to accepted quotes, a rerun registered beforehand reproduced 66 of 72 runs call for call and finished the other six: 69 of 72 runs finished, quality 1.64, +0.21 [0.09, 0.33] against dig then reason as evaluated (each judge alone above zero). On the final test (registered beforehand), the guards raise finished runs from 27 to 45 of 48 and halve refused calls (5.7 to 2.9 per run), but the quality gain is smaller and not clearly above zero: +0.12 [0.00, 0.25] (Claude Code +0.08 [−0.01, 0.19], `gpt-6-sol` +0.16 [0.01, 0.33]), most of it from one company. Unlike the advice-style aids of Section 12, these guards remove the failing action from what the model can produce, rather than telling the model about it.

**Evidence first.** With the guards in place, evidence judgement was the lowest judged dimension on the final test (Claude Code 0.46, `gpt-6-sol` 0.96 of 3). The judges' reasons were quotes counted as support for a theory they do not address, and company estimates labelled as independent evidence. Two more decoding-time changes were registered with adoption rules before they ran. *Evidence first* orders the assessment so that the quote and what it does not show come before the evidence kind and the direction (llama.cpp keeps the schema's property order in its grammar), and offers only `company_assertion` for a source the environment marks as a company claim. *Grounded beliefs* allows a 'likely' belief only for a theory that a real source the run weighed supports. On development, grounded beliefs changed the conclusions as designed (fewer unearned strong beliefs, ten recommendations to test first) but not their scores (−0.01 [−0.05, 0.03]). Evidence first passed its rule, and the arm with both was run once on each test set against the guarded default, judged blind by both judges (quality: default → evidence first, mean of judges, with the paired difference and 95% interval; the dimensions as Claude Code/`gpt-6-sol`):

| | Development (45) | Test set 2 (72) | Final test (48) |
|---|---|---|---|
| Quality | 1.56 → 1.63 | 1.64 → 1.67 | 1.56 → 1.62 |
| Difference | +0.07 [0.03, 0.11] | +0.03 [−0.00, 0.06] | +0.06 [0.00, 0.11] |
| Evidence judgement | 0.62/1.07 → 0.91/1.20 | 0.56/1.06 → 0.82/1.21 | 0.46/0.96 → 0.98/1.31 |
| Recommendation | 1.05/1.07 → 1.05/1.16 | 1.03/1.16 → 1.12/1.18 | 1.00/1.16 → 1.04/1.17 |

The gain is small on every set and sits in evidence judgement; the recommendation barely moves, so the weakness of the conclusion remains. Finished runs are 43 and 43, 69 and 65, 45 and 46. On test set 2 five runs that had finished stopped (three ran out of room in the final answer, one did not challenge its favoured theory, one reached the turn limit) and one that had stopped

finished; the same refusals also occur without the change. The mechanical numbers check fails more often (on test set 2 in 32 runs against 14), but runs with a number found nowhere, not even in the run's own test design, do not increase (9 to 8, 7 to 7, 3 to 2): the extra failures are recommendations that restate the run's own test thresholds. Under a rule fixed before the second judge scored test set 2, this is now part of the app's default.

# 12 What did not fix the remaining failures

After both tests we tried, on the development cases only, to repair version 1's remaining failures:

- invented numbers;
- weak or missing experiments;
- wrong evidence kinds;
- recommendations that do not follow the evidence;
- loops of refused calls.

Every attempt had its decision rule fixed before it ran. None succeeded. The failures fall into two different kinds, and we keep them apart.

## 12.1 Runtime interventions: the trained model does not use what it is told

Each of these changes the harness around the unchanged version 1 adapter. The evaluated runner and settings stay as they are.

- **Recovery controller.** We tested four aids: masking rereads, progress notes, step-budget awareness and citation repair. Budget awareness rescued 15 of 28 stuck checkpoints against 6 for the baseline, but most of the rescued finishes failed a check. End to end, over 45 paired runs, it changed 8 runs and left quality unchanged (+0.006 [−0.017, 0.031]). Most failed runs stop earlier, after four refusals in a row, where no late aid acts.
- **Evidence verification.** Before the final recommendation, the agent was shown which cited sources postdate the case, which numbers no cited source gives, and whether any weighed source supports its favoured theory. It could revise or answer "insufficient evidence". Over 45 paired runs it gave 19 unsupported recommendations against 20, the same validated completion, and no abstention. After 26 of 27 reviews the model immediately rewrote its text without new evidence work, and most findings survived.
- **Constrained, evidence-consistent position ("V1.1").** Constrained decoding offered as the favoured theory only those the agent's own weighings support. This collided with an existing guard: a theory must be challenged before it can be stated as the position. Finishes fell from 24 to 14 and validated completion from 3 to 1, so it was rejected under its fixed rule.
- **Novelty-first search ("V1-r").** The same keyword ranking, but each search returned records the run had not yet been shown. It doubled the sources surfaced (17.4 against 8.8 per run), but the agent read as many, weighed fewer (2.2 against 3.0), and weighed a contradicting source in the same 2 of 45 runs. Validated completion fell from 3 to 0, and the hits were less relevant. It was rejected under its fixed rule.

Advice does not change a 3B model's behaviour when nothing in its training data showed that behaviour. Constraints change behaviour, but they interact with the environment's other guards.

### 12.2 Training data: the teacher's errors are in the student

An audit of the 158 teacher investigations that trained version 1 found the teacher's own weaknesses:

- 74% of these 158 recommendations use a number from no source;
- 38% of its evidence kinds match the annotation, with `independent_study` often given to encyclopedia, news and company-history text;
- in 74 of its 109 historical cases (68%; 47% of all 158 investigations) the recommendation cites a source published after the case date, and in a blind content sample about half of those quotes reveal later outcomes.

Version 1 reproduces the first two: invented numbers in 21% and 31% of its finished recommendations on the two test sets, and kinds matching in 58% and 46%. The frozen test cases are present-day, so the dating problem cannot affect their scores. The dossier annotation itself agreed with a blind relabelling on 45 of 60 records, with 5 more defensible.

**Corrective distillation** tried to fix this with the same teacher. We took states where version 1 went wrong on training companies, asked the teacher for the right next step, and kept only answers that passed provenance, exact replay, split and family gates. Of 90 answers the gates accepted, an advisory substantive review rated 3 strong, and spot checks confirmed its reasons: later-dated sources, wrong kinds, overstated support, and tests that varied only the message. A structured format fixed the teacher's truncated test designs, but then it declined tests the cases warrant. We had fixed a threshold of 30 strong examples across three families, including experiment design, before training. Command A+ did not meet it. A hosted teacher (DeepSeek V4-Pro, with V4.1-Flash) then met it with 105 corrective rows, and version 2 was trained on top of version 1; on its development screen it finished 0 of 15 runs against version 1's 7, and it was stopped.

**What this does and does not show.** It shows that filtering cannot remove errors the teacher makes systematically, and that the student inherits them. It does not show that better training data would fail. A stronger or human-corrected teacher for evidence judgement and experiment design remains the most direct next step, and it was not tested.

## 13 aemon: the agent as a product

An agent that only exists in an evaluation harness is not useful to a manager. aemon runs the fine-tuned model (by default version 1 gathering the evidence and version 3 reasoning, with the loop guards of Section 9) locally (FastAPI and React over a llama.cpp server) and is designed, after a review of how current research assistants present long-running work, for readers who are not analysts.

- **Answer first.** A finished investigation opens with the recommendation, a confidence label and score, the next step and what would change the answer, then four headline numbers, then the explanations compared, the test, the sources, and only then the full trace.
- **Numbers counted from the record.** Every chart is computed from the investigation's own record, never estimated: how likely each explanation was judged before and after new evidence (an arrow plot on the five-step scale), how many weighed sources support or contradict each explanation (diverging bars), the mix of independent sources and company claims, and the test's cost and duration against its budget.
- **A confidence score out of five.** One point for each of five plain checks: the leading explanation is judged likely or better; it is judged very likely; at least two independent sources support it; no independent source contradicts it; at least three sources were weighed for it. Five is High, three or four Moderate, two or fewer Low, and a draft written after a

run stopped early is never more than Low. On the final test's 28 finished runs the score was 2 in 24 runs, 3 in two and 4 in two: the agent never judges its favourite very likely, and it weighs few independent sources, which is exactly what the iteration targets.

- **Follow-up questions grounded in the investigation.** A follow-up is answered by the base model (adapters off) from the investigation's record and source excerpts only, under a schema whose citations can name only sources in the record; "Re-investigate with this" reruns the investigation with the new information, files or links instead.
- **Steps a reader can follow.** The trace is shown as the seven steps a manager would recognise (understand the question, gather sources, list explanations, weigh the evidence, stress-test the favourite, design a test, recommend), each expandable to its actions and each action to exactly what was read, quoted or decided.
- **Investigations that do not die.** Runs keep going when the page is closed; every turn is saved, and after a restart unfinished runs resume by replaying their saved turns, each of which must reproduce its saved result. A run that stops early can be continued from where it stopped, and if it ends without a recommendation aemon writes a labelled draft from what it found.
- **Memory, projects, attachments and web search.** Notes about the person, the company or a project are given to the agent as context, never as evidence; attached files and links become quotable sources labelled as the person's own; an optional live web search saves every page it reads and labels it unchecked; a mind map shows every investigation; every result downloads as a PDF report.

None of these features is part of the evaluated setup; the reported results apply to the frozen configuration.

## 14 Limitations

- **Generalisation.** Four test companies, all business-to-business manufacturers asked the same family of question, support only a limited claim. The second test set addresses this partly; other industries and other kinds of strategic question remain untested.
- **Replications.** Three attempts per case at temperature 0 are close to repeats; the effective sample is nearer 16 cases than 48 runs, which is why the intervals resample whole cases. The cases are themselves 4 companies with 4 variants of one question, identical until the what-if arrives; resampling whole companies widens the final-test interval to $[0.63, 1.05]$ (bootstrap) or $[0.44, 1.23]$ (cluster-robust $t$, 3 degrees of freedom), and test set 2's to $[0.29, 0.75]$ or $[0.25, 0.78]$. With 4 companies any interval is rough; all 16 cases and all 4 companies are higher.
- **Judges.** Both judges are language models. They were qualified and agree closely on the order of runs, but not on the level (on the same fine-tuned runs, problem framing 1.62 from Claude and 2.90 from Codex), so the comparisons rest on the order and absolute scores mean little. They may share biases, and no strategy expert has scored the runs.
- **Absolute quality.** 1.48 of 3 is a large relative gain but a modest absolute score; 20 of 48 runs ended before a recommendation, and the recommendation, revision, experiment design and evidence judgement remain the weakest dimensions.
- **Comparators.** The comparison is with the same base model under its best prompt, which reads the full transcript: 25 of its 48 runs stop at the context limit, and a stage never reached scores 0. Comparing each pair only on the dimensions both runs were scored on gives +0.92 [0.83, 1.01] (test set 2: +0.90 [0.81, 0.99]), so the base model is weaker even

on the stages it reaches (`reports/judge/robustness.json`). The teacher was not run on the test sets for quality. Claude Haiku 5.5 was, after the frozen results, and scores higher than the final system ($-0.67$ and $-0.63$) at lower cost.

- **Beliefs and confidence.** Beliefs are the agent's own verbal scale, not calibrated probabilities, and the belief check tests direction only.
- **Sources and outcomes.** The dossier is fixed and curated, with short excerpts; what-if results are written by the case authors; no recommendation was checked against what a company later did, and no proposed test was run. The results measure the quality of the investigation, not whether its advice would pay off.
- **Training.** One training run and one seed; variance across runs is not measured. The training data has a known defect, the 240-character cut of test designs, and no example of declining a test.
- **Controls.** The scripted negative controls could not run on the held-out cases and are reported for development only.
- **Teacher errors in the training data.** The rows are valid, accepted calls, but their content carries the teacher's errors (Section 12): invented numbers, wrong evidence kinds, overstated support, and, in historical training cases, sources published after the case date. The model reproduces the first two.
- **Our own instruments.** The mechanical experiment check is lenient: on development it passes 9 tests, 8 of which fail stricter design screens. The evidence-kind annotation matches a blind relabelling on 75% of records, 83% counting defensible alternatives. Both are reported as they are, and the frozen numbers use them unchanged.
- **Mechanical validity.** Few runs pass every check: 1 of 48 on the final test and 4 of 72 on the second. The quality gains are measured by blind judges, not by mechanical validity.

## 15 Conclusion

A small open model can learn a meaningful part of a disciplined strategy procedure from a larger model's tool-using investigations, provided the environment makes the procedure executable and the data are filtered for exactness. On a frozen test it framed problems better, built genuinely distinct causal theories, gathered and weighed quoted evidence and finished far more investigations than its base model. A closed small model (Claude Haiku 5.5) still does better, and for less. It did not learn what its teacher rarely showed it: declining a test that is not worth its cost, responding to directional new evidence, and writing complete decision rules, and it inherited the teacher's habits of unsupported numbers and loose evidence labels. Neither runtime aids nor corrective examples from the same teacher repaired these, which places the remaining gap in the quality of the supervision. A better teacher for evidence judgement is the next step. Equally important is how the result was obtained: an environment that refuses rather than trusts, training data bound to its provenance, judges qualified before use, and a test frozen and run once.

## Team and acknowledgements

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# A Reproducibility

- **Model.** Tiny Aya Global, revision af89d21; adapter sha256 740c4a9b…; evaluated GGUF adapter sha256 41794b57…, byte-identical across three conversions and to the file gated.
- **Teacher.** Command A+ (05-2026), Q4_K_M GGUF in four parts, first part sha256 03cf0669…; llama.cpp, 4 slots of 16,384 tokens, reasoning budget 512 from 4 November.
- **Final test freeze.** Case digest dce26d4f…; runner code d3aa608; job script 0377eae; llama.cpp context 12,288, 520 tokens per turn, 45 turns, 4 refusals in a row; base prompt `staged`, fine-tuned prompt `compact`; judge configuration judge-v1-two-backends-2; bootstrap 4,000 resamples, seed 20261106.
- **Code and data.** The environment, pipeline, training and evaluation code and the job scripts are in the public repository; the model, the dossier, the cases, the training rows, every evaluation run, the judge packets, both judges' scores and the reports are on Hugging Face, with a manifest of the sha256 of every private original and its published copy. Since 11 November the public dossier is byte-identical to the one the evaluated runs read.